

## WEEK-7 SKILL — Student Records CRUD API [Sampath, 2520090005]

### POST:

REST client interface showing a POST request to `http://localhost:3000/students`. The request body is a JSON object:

```
{
  "name": "Sampath",
  "age": 18,
  "course": "CSIT",
  "email": "sampath@klh.edu.in"
}
```

The response status is **201 Created**. The response body is a JSON object:

```
{
  "name": "Sampath",
  "age": 18,
  "course": "CSIT",
  "email": "sampath@klh.edu.in",
  "_id": "6ab0d6e94bd72bb2efd336e4",
  "createdAt": "2026-09-21T07:04:09.524Z",
  "updatedAt": "2026-09-21T07:04:09.524Z",
  "__v": 0
}
```

### After POST in mongosh:

```
student_records> db.students.find()
[
  {
    _id: ObjectId('6ab0d6e94bd72bb2efd336e4'),
    name: name: Sampath,
    age: 18,
    course: 'CSIT',
    email: email: sampath@klh.edu.in,
    createdAt: ISODate('2026-09-21T07:04:09.524Z'),
    updatedAt: ISODate('2026-09-21T07:04:09.524Z'),
    __v: 0
  }
]
```

### GET all data:

REST client interface showing a GET request to `http://localhost:3000/students`. The response status is **200 OK**. The response body is a JSON array containing two student records:

```
[
  {
    "_id": "6ab0d6e94bd72bb2efd336e4",
    "name": "Sampath",
    "age": 18,
    "course": "CSIT",
    "email": "sampath@klh.edu.in",
    "createdAt": "2026-09-21T07:04:09.524Z",
    "updatedAt": "2026-09-21T07:04:09.524Z",
    "__v": 0
  },
  {
    "_id": "6ab0d7b74bd72bb2efd336e5",
    "name": "K Venkat Ram Reddy",
    "age": 23,
    "course": "MS in BA",
    "email": "kvrreddy@gmail.com",
    "createdAt": "2026-09-21T07:07:35.746Z",
    "updatedAt": "2026-09-21T07:07:35.746Z",
    "__v": 0
  }
]
```

## GET all data in mongosh:

```
student_records> db.students.find()
[
  {
    _id: ObjectId('6ab0d6e94bd72bb2efd336e4'),
    name: 'Sampath',
    age: 18,
    course: 'CSIT',
    email: 'sampath@klh.edu.in',
    createdAt: ISODate('2026-09-21T07:04:09.524Z'),
    updatedAt: ISODate('2026-09-21T07:04:09.524Z'),
    __v: 0
  },
  {
    _id: ObjectId('6ab0d7b74bd72bb2efd336e5'),
    name: 'K Venkat Ram Reddy',
    age: 23,
    course: 'MS in BA',
    email: 'kvrreddy@gmail.com',
    createdAt: ISODate('2026-09-21T07:07:35.746Z'),
    updatedAt: ISODate('2026-09-21T07:07:35.746Z'),
    __v: 0
  }
]
```

## GET one with specific id:

The screenshot shows a REST client interface with a GET request to `http://localhost:3000/students/6ab0d6e94bd72bb2efd336e4`. The response is a JSON object representing a student record with the following fields: `_id`, `name` ("Sampath"), `age` (18), `course` ("CSIT"), `email` ("sampath@klh.edu.in"), `createdAt`, `updatedAt`, and `__v` (0). The status is 200 OK.

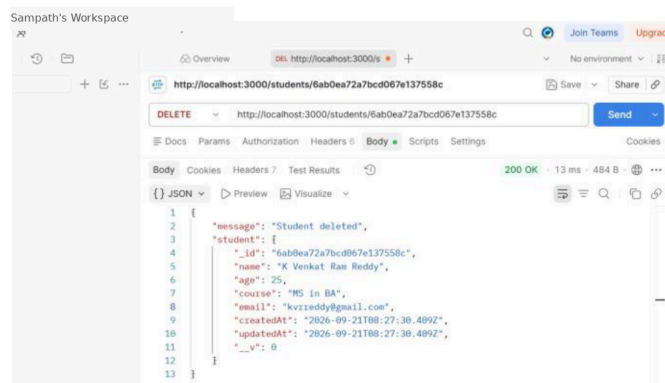
## PATCH → changed age from 23 to 25:

The screenshot shows a REST client interface with a PATCH request to `http://localhost:3000/students/6ab0d7b74bd72bb2efd336e5`. The request body is a JSON object with the field `age` set to 25. The response is a JSON object representing the updated student record, where the `age` is now 25 and the `updatedAt` timestamp has changed. The status is 200 OK.

After modification of age output in mongosh:

```
{
  _id: ObjectId('6ab0ea72a7bcd067e137558c'),
  name: 'K Venkat Ram Reddy',
  age: 25,
  course: 'MS in BA',
  email: 'kvrreddy@gmail.com',
  createdAt: ISODate('2026-09-21T08:27:30.409Z'),
  updatedAt: ISODate('2026-09-21T08:27:30.409Z'),
  __v: 0
}
```

**DELETE:**



Observation of output in mongosh:

```
student_records> db.students.find()
[
  {
    _id: ObjectId('6ab0d6e94bd72bb2efd336e4'),
    name: Sampath
    age: 18,
    course: 'CSIT',
    email: sampath@kjh.edu.in
    createdAt: ISODate('2026-09-21T07:04:09.524Z'),
    updatedAt: ISODate('2026-09-21T07:04:09.524Z'),
    __v: 0
  }
]
student_records>
```

**Folder Structure:**

Week\_7\_StudentRecordsAPI

- ├── controllers
  - └── studentController.js
- ├── models
  - └── Student.js
- ├── routes
  - └── studentRoutes.js
- └── server.js

└─ package.json

**Code:**

**package.json**

```
{
  "name": "week_7_studentrecordsapi",
  "version": "1.0.0",
  "type": "module",
  "description": "",
  "main": "server.js",
  "scripts": {
    "start": "node server.js",
    "dev": "nodemon server.js"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "dependencies": {
    "express": "^5.2.1",
    "mongoose": "^9.10.1"
  },
  "devDependencies": {
    "nodemon": "^3.1.14"
  }
}
```

**models/Student.js**

```
import mongoose from 'mongoose';

const studentSchema = new mongoose.Schema({
  name: {
    type: String,
    required: true,
    trim: true
  },
  age: {
    type: Number,
    required: true,
    min: 16,
    max: 100
  },
  course: {
    type: String,
    required: true,
    trim: true
  },
  email: {
    type: String,
```

```

        required: true,
        unique: true,
        lowercase: true
    }
}, {
    timestamps: true
});

const Student = mongoose.model('Student', studentSchema);

export default Student;

```

### controllers/studentController.js

```

import Student from '../models/Student.js';

export const createStudent = async (req, res) => {
    try {
        const student = await Student.create(req.body);
        res.status(201).json(student);
    } catch (error) {
        res.status(400).json({ error: error.message });
    }
};

export const getAllStudents = async (req, res) => {
    try {
        const students = await Student.find();
        res.status(200).json(students);
    } catch (error) {
        res.status(500).json({ error: error.message });
    }
};

export const getStudentById = async (req, res) => {
    try {
        const student = await Student.findById(req.params.id);

        if (!student)
            return res.status(404).json({ message: 'Student not found' });

        res.status(200).json(student);
    } catch (error) {
        res.status(400).json({ error: error.message });
    }
};

export const updateStudent = async (req, res) => {
    try {

```

```

const student = await Student.findByIdAndUpdate(
  req.params.id,
  req.body,
  { new: true, runValidators: true }
);

if (!student)
  return res.status(404).json({ message: 'Student not found' });

res.status(200).json(student);
} catch (error) {
  res.status(400).json({ error: error.message });
}
};

export const deleteStudent = async (req, res) => {
  try {
    const student = await Student.findByIdAndDelete(req.params.id);

    if (!student)
      return res.status(404).json({ message: 'Student not found' });

    res.status(200).json({
      message: 'Student deleted',
      student
    });
  } catch (error) {
    res.status(400).json({ error: error.message });
  }
};

```

### routes/studentRoutes.js

```

import express from 'express';

import {
  createStudent,
  getAllStudents,
  getStudentById,
  updateStudent,
  deleteStudent
} from '../controllers/studentController.js';

const router = express.Router();

router.post('/', createStudent);
router.get('/', getAllStudents);
router.get('/:id', getStudentById);

```

```
router.patch('/:id', updateStudent);
router.delete('/:id', deleteStudent);
```

```
export default router;
```

## server.js

```
import express from 'express';
import mongoose from 'mongoose';
import studentRoutes from './routes/studentRoutes.js';

const app = express();
const PORT = 3000;

app.use(express.json());

app.get('/', (req, res) => {
  res.json({
    message: 'Student Records API is running'
  });
});

app.use('/students', studentRoutes);

mongoose.connect('mongodb://127.0.0.1:27017/student_records')
  .then(() => {
    console.log('MongoDB connected');

    app.listen(PORT, () => {
      console.log(`Server running on http://localhost:${PORT}`);
    });
  })
  .catch((error) => {
    console.error('MongoDB connection failed:', error.message);
  });
```